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Studierichting: Informatica
Oefeningenreeks 3D nr. 2.6

$f(x) = 3x^4 + 8x^3 - 30x^2 - 72x$

$f'(x) = 12x^3 + 24x^2 - 60x - 72$

$f'(x) = 0 \Leftrightarrow 12x^3 + 24x^2 - 60x - 72 = 0$

$\Leftrightarrow x^3 + 2x^2 - 5x - 6 = 0$

$$\begin{array}{r|rrrr} -1 & 1 & 2 & -5 & -6 \\ & & -1 & -1 & 6 \\ \hline & 1 & 1 & -6 & 0 \end{array}$$

$x^2 + x - 6 = 0$

$\Leftrightarrow (x+3)(x-2) = 0$

$\Leftrightarrow x = -3$ of $x = 2$

x	$-\infty$	-3		-1		2	$+\infty$
$f'(x)$	$-$	0	$+$	0	$-$	0	$+$
$f(x)$	$\searrow$	m	$\nearrow$	M	$\searrow$	m	$\nearrow$
		-27		37		-152	

References